

Neural Operators: Update

From what we have to what we do next

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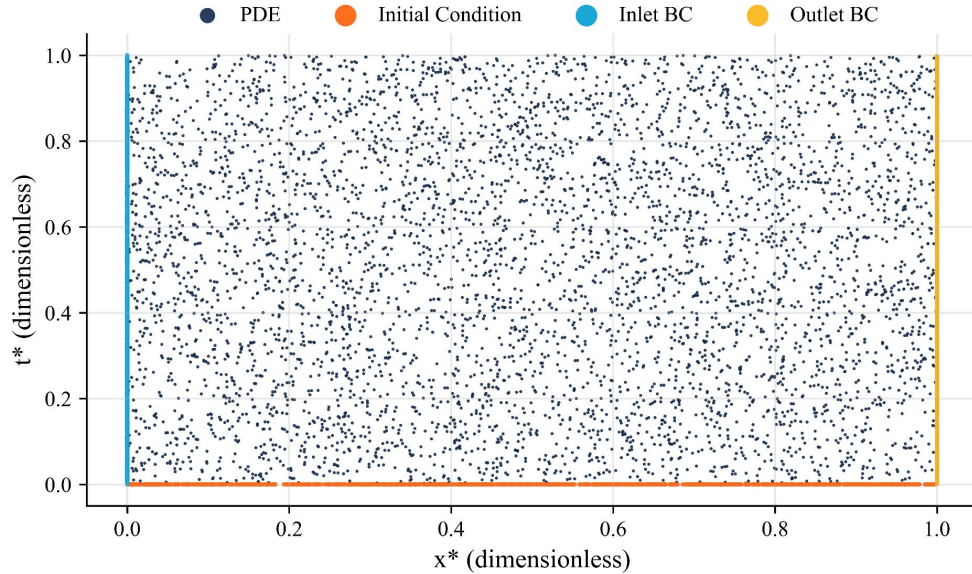
Review

What we currently have

Baseline PINN:

a small fully connected MLP: $2 \rightarrow 12 \rightarrow 12 \rightarrow 12 \rightarrow 1$, with Tanh in hidden layers and Sigmoid on the output, trained via physics-informed loss.

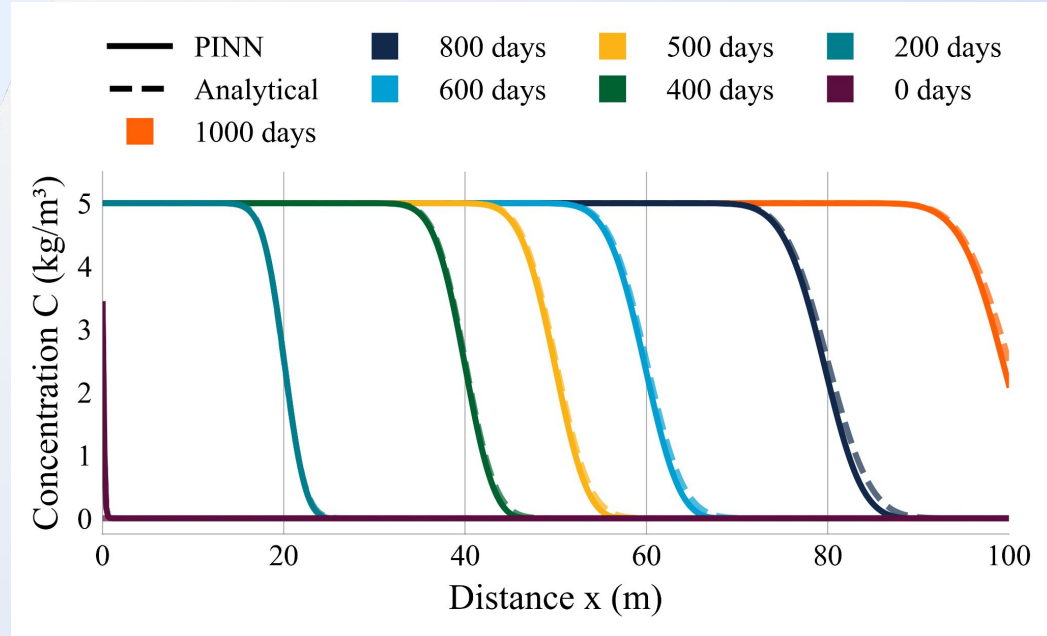
$$(x^*, t^*) \mapsto C^*$$



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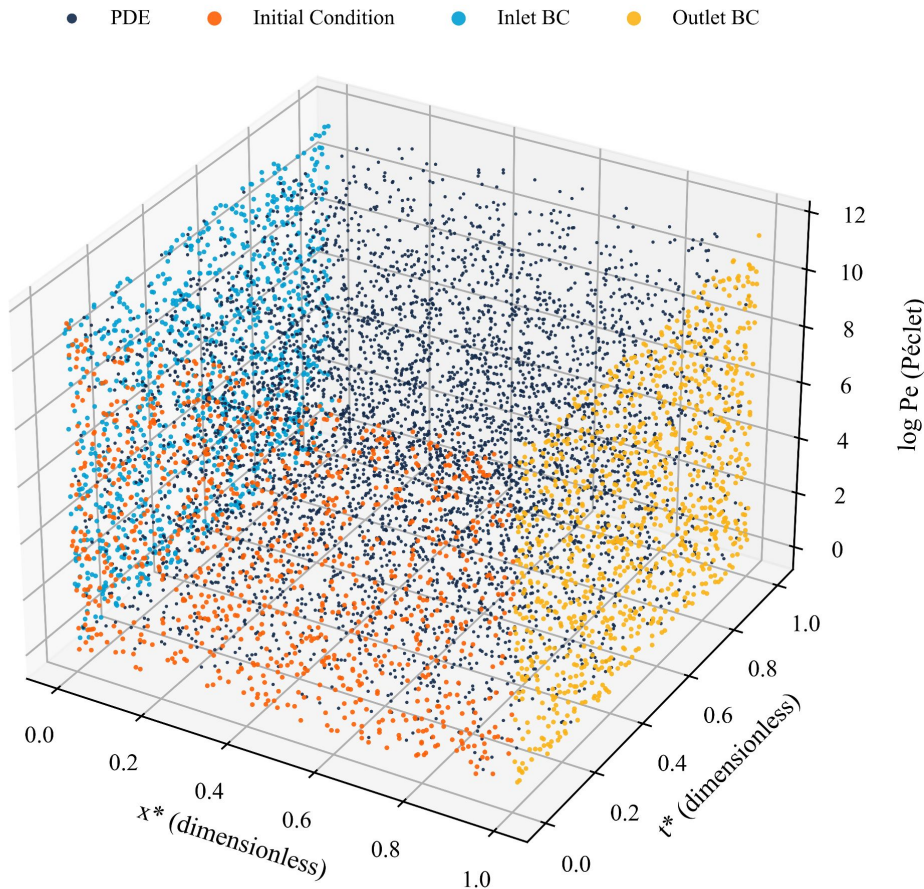
$$(x^*, t^*) \mapsto C^*$$



Parametric PINN:

A small fully connected MLP: $3 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 1$, with Tanh in hidden layers and Sigmoid on the output, trained via physics-informed loss.

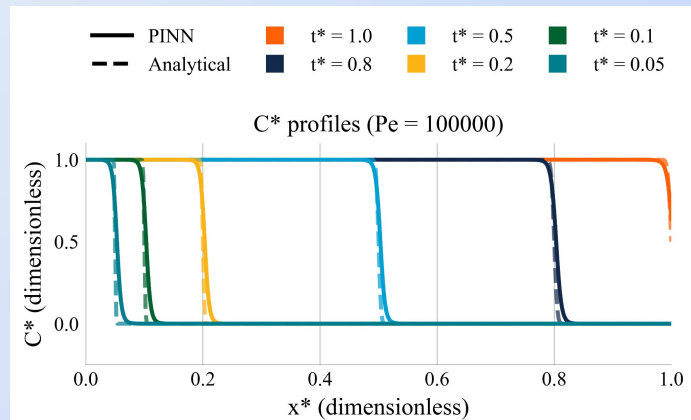
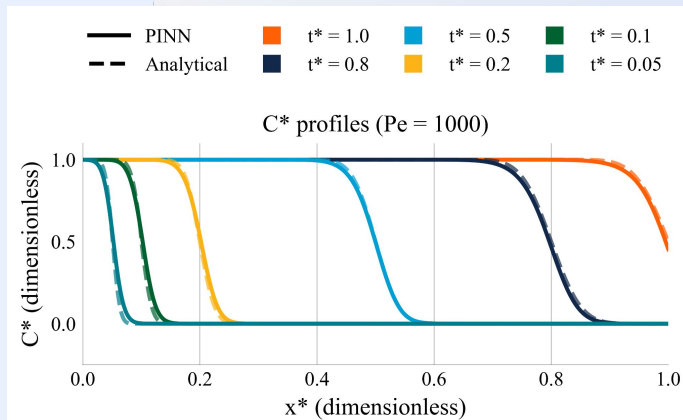
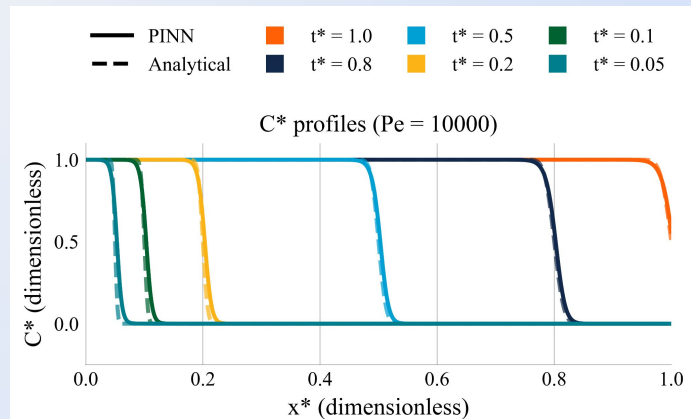
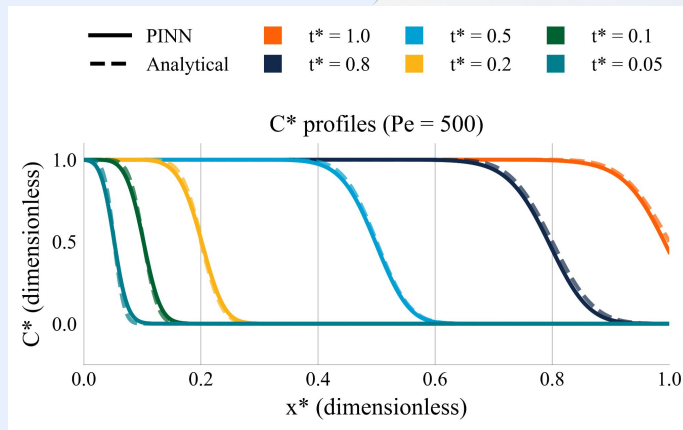
$$(x^*, t^*, \log Pe) \mapsto C^*$$



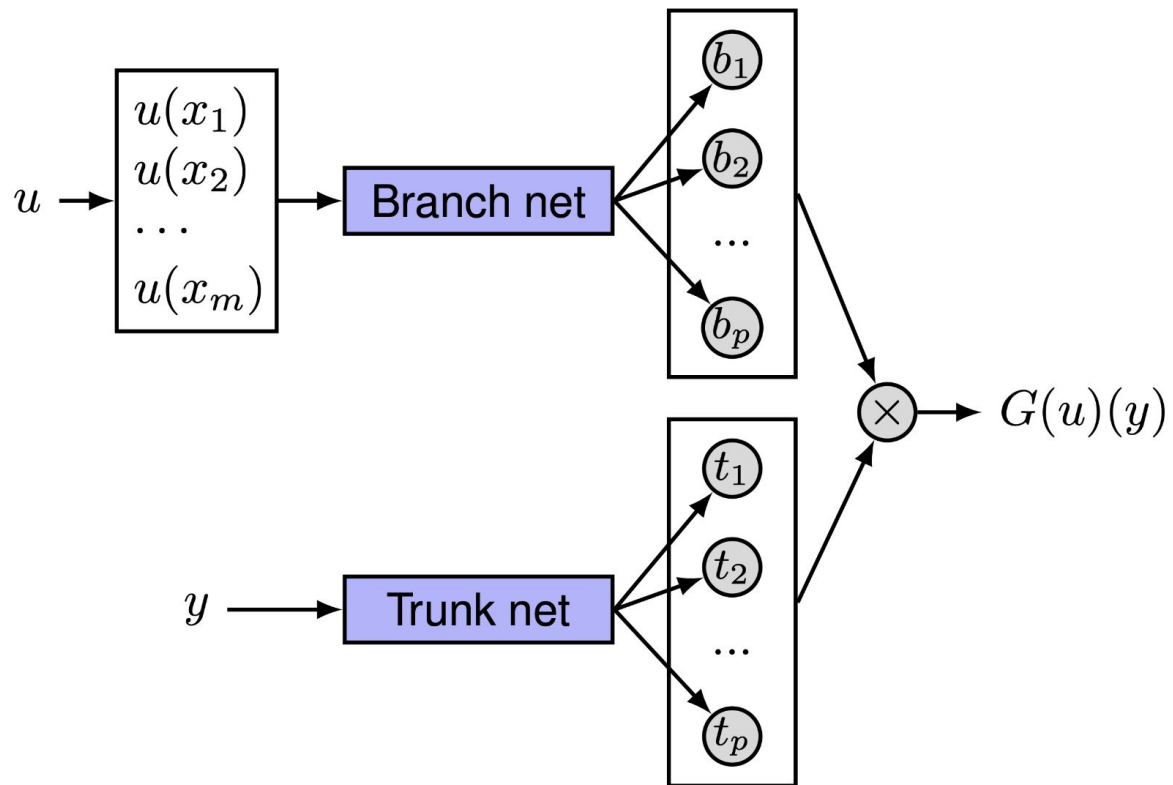
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$$(x^*, t^*, \log Pe) \mapsto C^*$$



Neural Operator



Neural Operator

**Marwan's question on
what the input to the DeepONet is**

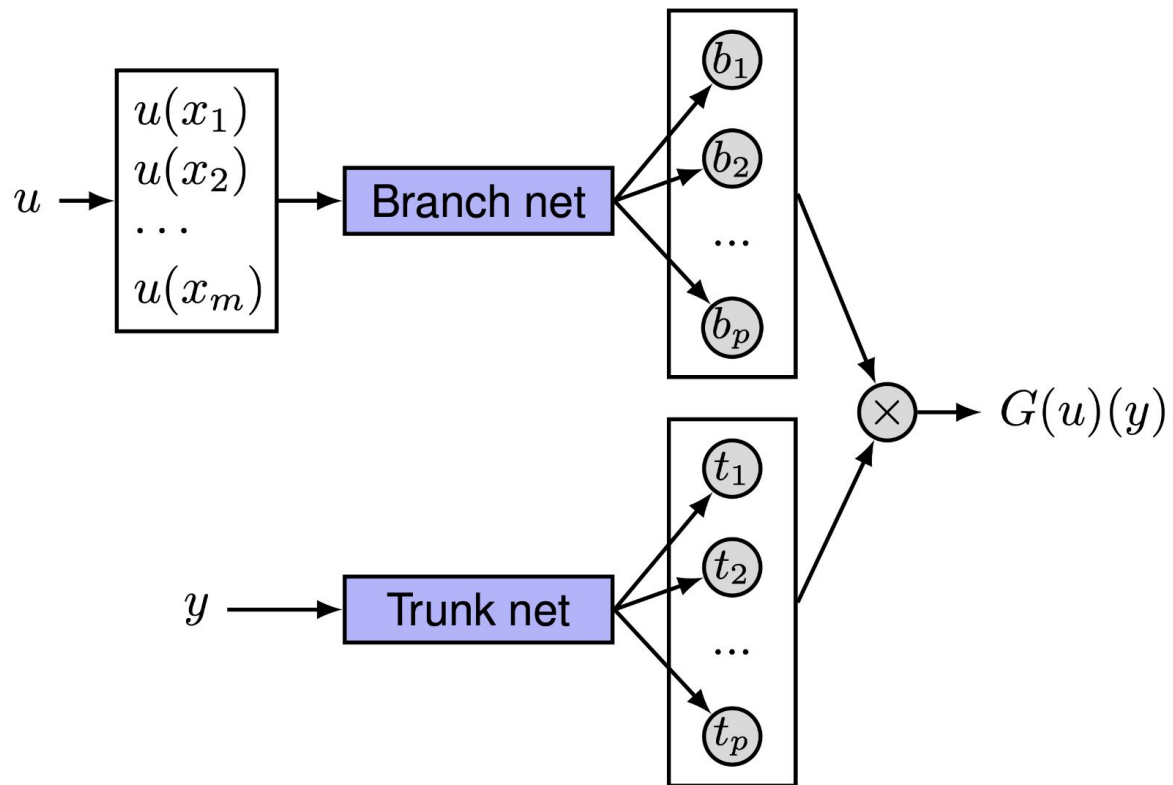
Sensors and queries

$$\mathbf{u}_i = \begin{bmatrix} u_i(x_1) \\ \vdots \\ u_i(x_m) \end{bmatrix} \in \mathbb{R}^m.$$

Input sampling (sensors)

$$\mathbf{v}_i = \begin{bmatrix} v_i(y_1) \\ \vdots \\ v_i(y_q) \end{bmatrix} \in \mathbb{R}^q.$$

Output evaluation (queries)

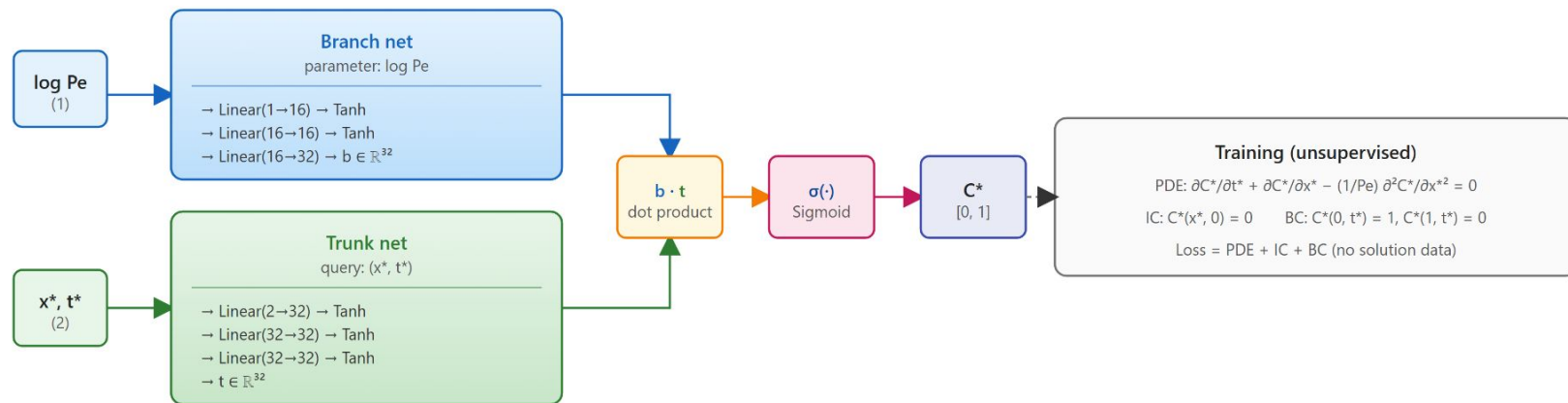


Neural Operator

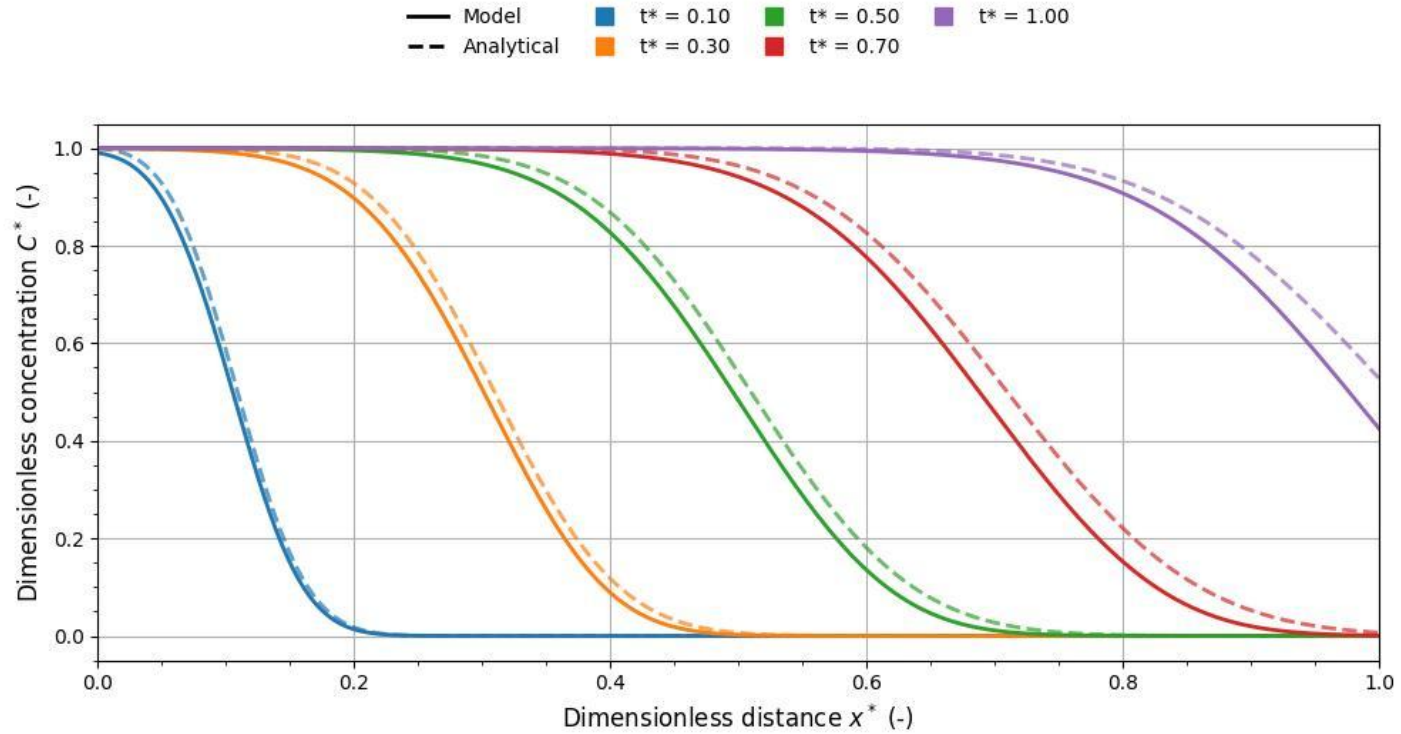
Implementation on our problem

DeepONet neural operator (unsupervised)

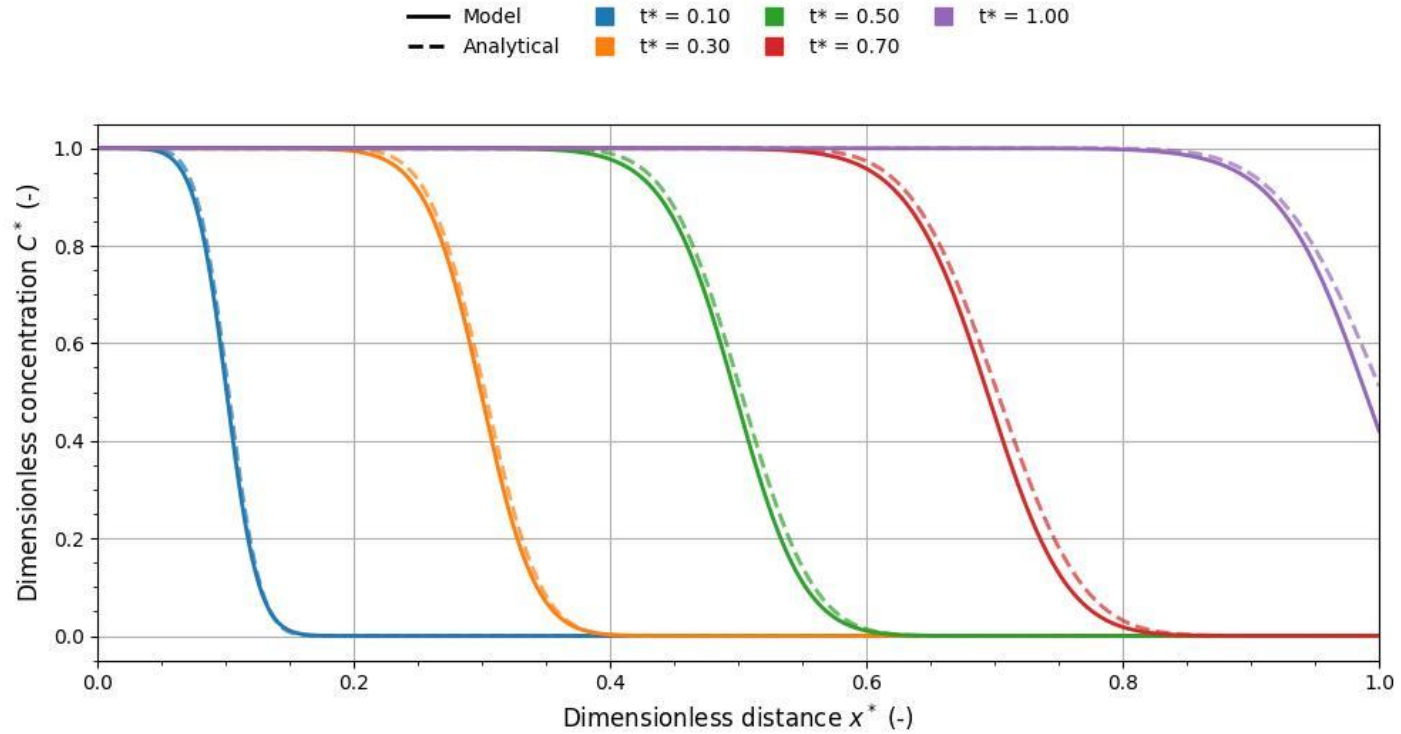
1D advection–dispersion: $C^* = f(x^*, t^*, \log Pe)$



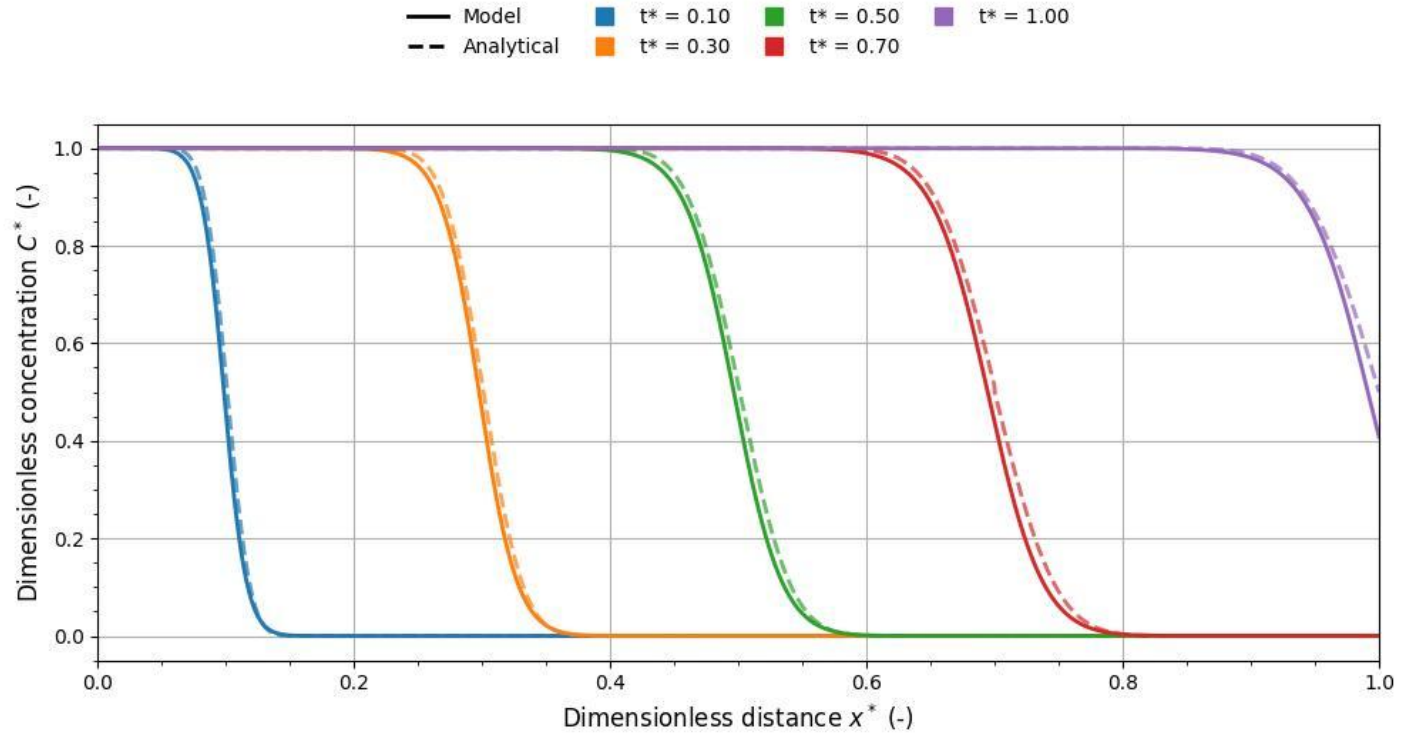
DeepONet (unsupervised): C^* vs x^* ($Pe = 100$)



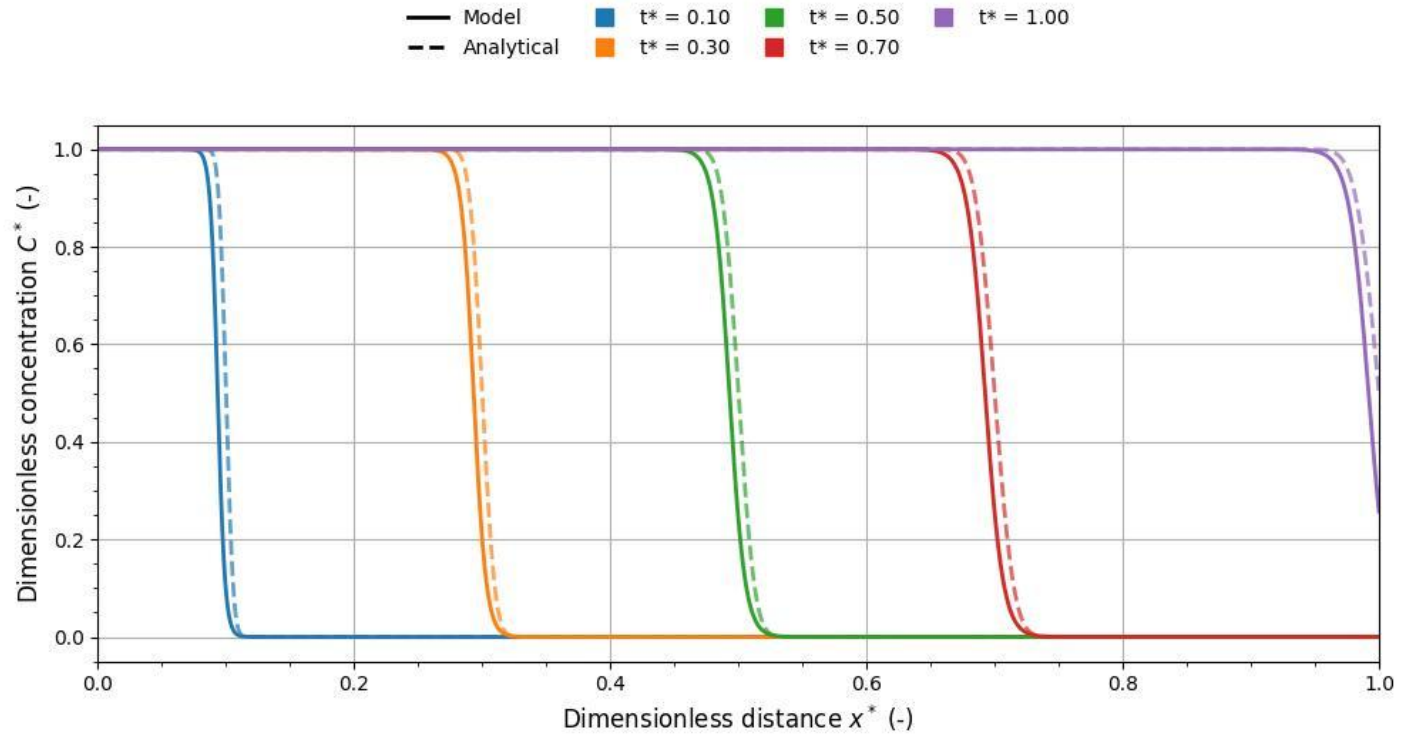
DeepONet (unsupervised): C^* vs x^* ($Pe = 500$)



DeepONet (unsupervised): C^* vs x^* ($Pe = 1000$)



DeepONet (unsupervised): C^* vs x^* (Pe = 10000)



Possible trajectories

What problem to focus on?

One Potential Option

Physics-guided neural operator mapping heterogeneous aquifer properties and source conditions to spatiotemporal contaminant plumes for 2D reactive advection-diffusion in saturated groundwater.

Inputs

- Heterogeneous hydraulic conductivity field $K(x,y)$ or velocity field $q(x,y)$
- Source/Leak location and time history $s(x,y,t)$
- Boundary heads / boundary masks

Outputs

- Full concentration field $C(x,y,t)$ at multiple times
- Breakthrough curves at monitoring wells



Thank You!